



In search of a learning based storm aware ionospheric delay prediction model for near Equatorial Ionization Anomaly crest

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Abstract

Ionospheric scintillation over near Equatorial Ionization Anomaly crest location, Bangalore is predicted during G2, G3 and G5 geomagnetic storm that happened on May, 2024. LSTM and Transformer is used for the storm aware forecasting model development. Statistical parameters like RMSE, Correlation coefficient has been calculated for amplitude scintillation (S4) prediction. Contingency parameters like F1 score, POD, FAR, CSI have also calculated. The study will be helpful to choose an appropriate algorithm as an input of early warning system development in case of accidental positioning.

1. Introduction

Understanding of the delayed response of ionosphere during the disturbed period is a subject of interest as ionospheric delay causes a significant error in satellite-based navigation. TEC of disturbed ionosphere is the prime cause of delay and the correlation of TEC with solar extreme ultraviolet has been studied in literature [1]. VTEC prediction during solar maxima period over near Equatorial Ionization Anomaly (EIA) crest, Bangalore (13.02°N, 77.57°E) has been done using multi-input Long Short-Term Memory (LSTM) [2]. Performance assessment of the model has done during disturbed and quiet space weather and compared with physics based conventional model. The performance statistic shows superiority of learning-based model. LSTM based prediction has been done over Bangaluru [3] and compare with a convectional physics based model. The deep learning model outperforms the traditional model. The delayed response of ionosphere over the location Bhopal (23.2°N, 77.6°E) near EIA crest has been studied during solar minima period also [4]. [5] uses hybrid nonlinear autoregressive neural network with external input (NARX) for the prediction of TEC over two near EIA stations Hyderabad (17.41°N, 78.55°E) and Bangalore. Hybrid LSTM-CNN is used for TEC prediction over multiple stations, Bangalore (13.02°N, 77.57°E), Guntur (16.44°N, 80.62°E), and Lucknow (26.91°N, 80.95°E) near EIA crest [6]. Most of the discussed model in trained either with the observation of a complete solar cycle or with the observation of minima time. Therefore, the model faces challenge while experiencing EIA

intensification due to disturbance dynamo electric fields prompt penetration electric field during severe ionospheric storm. Therefore, the RMSE provided by existing model during all weather conditions is not sufficient to describe storm condition. Therefore, a dedicated storm aware learning based model has been proposed after comparison of two deep learning models.

2. Data Collection and Processing Methodology

TEC data has been collected from the International GNSS Service network (<https://igs.org/data/>) over near EIA locations, Bangalore (13.02°N, 77.57°E) Fig. 1. Work flow diagram is shown in Fig. 2. Data pre-processing has been done considering sorting, normalization. Ionodelay(d_{iono}) is calculated as in equation 1.

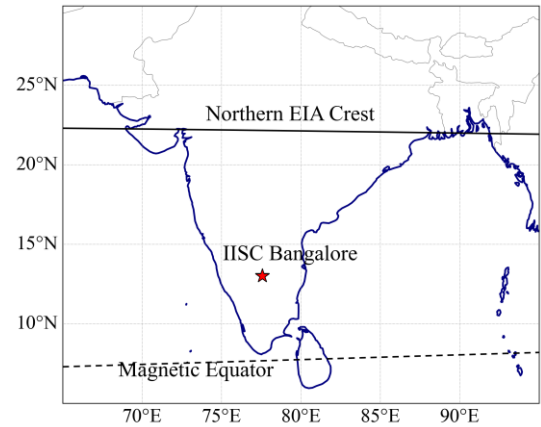


Figure 1. Study Location.

$$d_{iono} = \frac{40.3 \times 10^{16} \times TEC_{TECU}}{f^2} \quad (1)$$

Where, f = frequency

The observation has been for entire period of the year 2024. Deep learning-based delay prediction has been done using Transformer and LSTM over Bangalore historical delay data of minute resolution. The model is trained for 334 days

except May 2024. The validation has been done for 2nd May 2024 (G3 storm), 6th May (G2 storm) and 9th May (G5 storm). The configuration of computing facility used for the analysis NVIDIA GeForce RTX 3060 12 GB GPU with 64 GB RAM and AMD Ryzen 5 7600X processor. Datasets from IISc Bangalore was fed PyTorch-based LSTM and Transformer models individually to find storm time ionospheric delay.

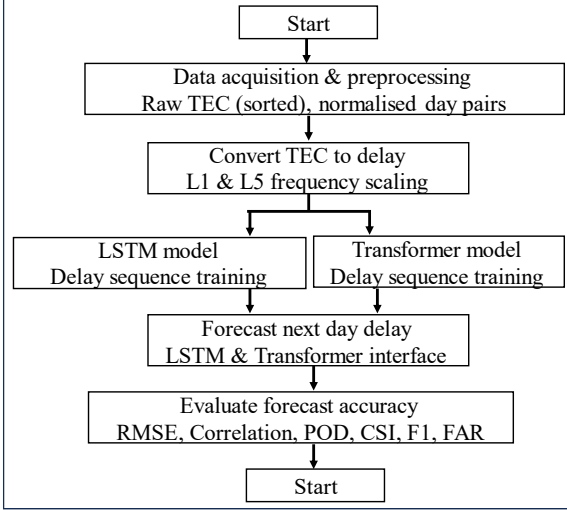


Figure 2. Work Flow Diagram.

3. Space Weather Condition During Storm

Fig. 3 shows the comparative analysis between the geomagnetic storm days (G2, G3, G5) and the preceding quiet days. Storm days (2nd May, 6th May and 10th May) and quiet days (1st May, 5th May and 9th May) are characterised by Dst, Kp and Ap. The solar parameters are collected from <https://www.spaceweatherlive.com/> & https://wdc.kugi.kyotou.ac.jp/dst_provisional/index.html accessed on (16th July 2026). Table I highlights the geomagnetic storm days on the basis of solar parameters.

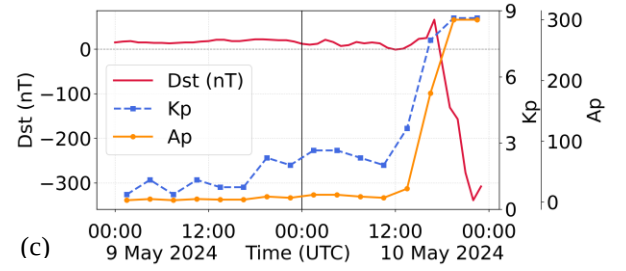
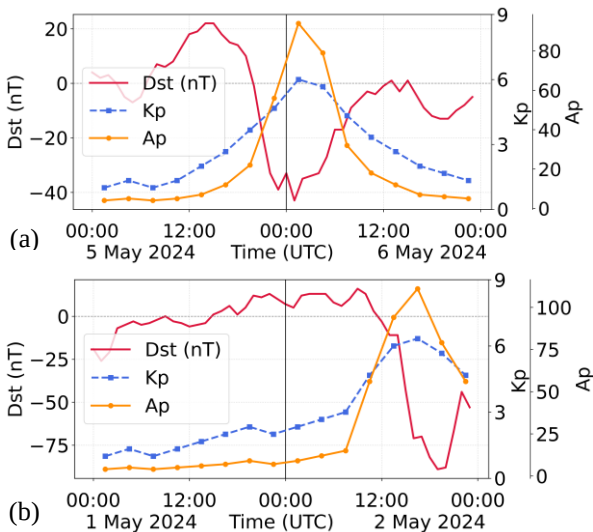


Figure 3. Solar parameter variation across three distinct event intervals (a) 5th May vs 6th May (G2), (b) 1st May vs 2nd May (G3), (c) 9th May vs 10th May (G5)

Table I. Classification of geomagnetic storms based on solar parameters.

Storm days	Dst	Kp	Ap
6 th May (G2)	-43	2.33	9
2 nd May (G3)	-89	6.67	111
10 th May (G5)	-339	8.67	300

4. Results

Ionodelay is predicted for both L1 and L5 frequencies. The prediction and observation is shown in Fig. 4. The performance of each of the model calculated using statical as well as contingency metrics.

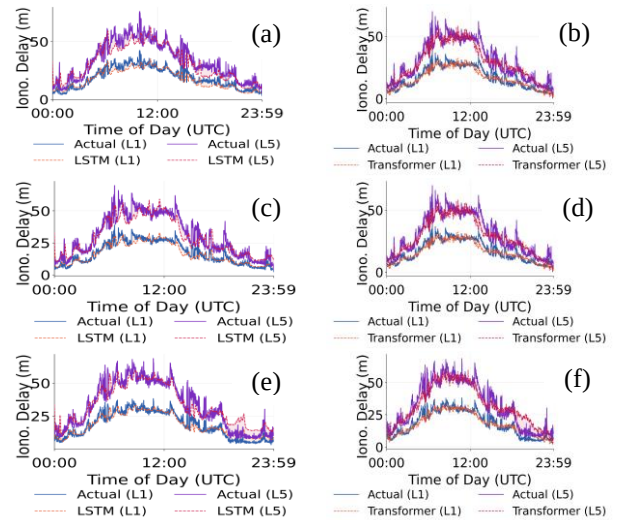


Figure 4. Temporal variation of Observed vs predicted model for GNSS L1 & L5 signal on 6th May (a) for LSTM (b) for Transformer, 2nd May (c) for LSTM (d) Transformer and 10th May for (e) LSTM (f) Transformer.

Table II, III and IV show the performance of each of the models for a comparatively weaker G2 storm, G3 storm and G5 storm day respectively. G3 shows a moderate storm strength, whereas G5 storm is the strongest storm that

happens within the last decade. Correlation coefficient and RMSE between the observed and predicted delay shows a comparatively better performance of prediction by Transformer model even for a strongest storm day. Contingency parameters like CSI, F1 score, POD and FAR is calculated [7] considering the threshold delay of 3 meter, 8 meter and 15 meter for all the storm days. For strongest storm day the competitive performance of both the methods are obtained showing a better tendency towards Transformer. For weak and moderate storm day performance of both the models are same.

Table II. (a) Performance (CSI, F1 score, POD, FAR), (b) RMSE and correlation metrics for 6th May.

(a)

	Freq (Hz)	Event Threshold (m)	CSI	F1 score	POD	FAR
LSTM	L1	3	1.00	1.00	1.00	0.00
		8	0.92	0.96	0.97	0.05
		15	0.82	0.90	0.84	0.02
	L5	3	1.00	1.00	1.00	0.00
		8	0.99	1.00	1.00	0.01
		15	0.93	0.96	0.97	0.04
Transformer	L1	3	0.98	0.99	0.98	0.00
		8	0.93	0.96	0.95	0.02
		15	0.83	0.91	0.85	0.03
	L5	3	1.00	1.00	1.00	0.00
		8	0.95	0.97	0.96	0.01
		15	0.94	0.97	0.96	0.02

(b)

Freq (Hz)	RMSE		Correlation Coefficient	
	LSTM	Transformer	LSTM	Transformer
L1	22.25	20.75	0.83	0.94
L5	22.95	20.00	0.83	0.94

Table III. (a) Performance (CSI, F1 score, POD, FAR), (b) RMSE and correlation metrics for 2th May.

(a)

	Freq (Hz)	Event Threshold (m)	CSI	F1 score	P O D	FAR
LSTM	L1	3	0.99	0.99	1.00	0.00
		8	0.91	0.95	0.98	0.07
		15	0.82	0.90	0.88	0.07
	L5	3	1.00	1.00	1.00	0.00
		8	0.98	0.99	1.00	0.01
		15	0.90	0.95	0.07	0.07
Transformer	L1	3	0.97	0.98	0.97	0.00
		8	0.92	0.96	0.96	0.04
		15	0.86	0.92	0.93	0.07

L5	3	0.99	0.99	0.99	0.00
	8	0.94	0.97	0.94	0.00
	15	0.92	0.95	0.96	0.04

(b)

Freq (Hz)	RMSE		Correlation Coefficient	
	LSTM	Transformer	LSTM	Transformer
L1	19.97	19.83	0.83	0.93
L5	20.00	19.23	0.83	0.93

Table IV. (a) Performance (CSI, F1 score, POD, FAR), (b) RMSE and correlation metrics for 10th May.

(a)

	Freq (Hz)	Event Threshold (m)	CSI	F1 score	POD	FAR
LSTM	L1	3	0.99	0.99	0.99	0.00
		8	0.74	0.71	0.76	0.12
		15	0.87	0.93	0.96	0.09
	L5	3	0.79	0.79	0.79	0.02
		8	0.96	0.98	0.96	0.02
		15	0.75	0.72	0.76	0.11
Transformer	L1	3	1.00	1.00	1.00	0.00
		8	0.80	0.89	0.97	0.17
		15	0.86	0.92	0.92	0.06
	L5	3	1.00	1.00	1.00	0.00
		8	0.99	0.99	1.00	0.00
		15	0.83	0.91	0.97	0.14

(b)

Freq (Hz)	RMSE		Correlation Coefficient	
	LSTM	Transformer	LSTM	Transformer
L1	3.52	3.63	0.83	0.92
L5	6.32	6.51	0.83	0.92

5. Discussion and Conclusion

The work was the advantage of robust, long range dependency tracking for the prediction of ionospheric delay using Transformer. The prediction using LSTM has also been compared. The complex, intricate relation during G5 geomagnetic storm is better identified using Transformer. Addition of more feature from space weather will make the proposed model more robust. The performance of contingency parameter shows an indication to better use the Transformer as an i/p of an early warning system to issue warning in case of accidental positioning.

6. Acknowledgements

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7. References

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